

12 Nitrogen and sulfur

AS · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Ammonia's lone pair makes it a base and ligand. The Haber process balances equilibrium yield with rate and cost using moderate conditions and a catalyst; catalyst does not shift equilibrium. Sulfur dioxide is an acidic, reducing oxide and an atmospheric pollutant; sulfuric acid manufacture involves reversible oxidation to SO_3 .

Worked example: follow the reasoning

$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ is exothermic, so lower temperature favours ammonia at equilibrium, though too low a temperature slows attainment.

Quick check · multiple choice

A What does an industrial catalyst change in Haber equilibrium? [1]

A Position of equilibrium

B K_c

C Enthalpy change

D Rate of attainment

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Explain why industrial ammonia production uses a catalyst and recycles unreacted gases. [3]

Answers and explanation

Quick check: D. Rate of attainment

Question 1 [3 marks]

Catalyst speeds forward and reverse reactions [1]; does not alter equilibrium yield [1]; recycling unreacted N_2 and H_2 raises overall conversion [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

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